BOUNDARY / PROJECT REFERENCE

Libraries and SDKs

Java Android app + Vercel API + Neon PostgreSQL. Short notes based on this repository, 16 October 2026.

- **AndroidX AppCompat 1.8.0** - AppCompatActivity provides the base class for the app's screens and theme compatibility.
- **AndroidX Core / ViewCompat** - Applies system-bar insets so controls remain clear of the status and navigation bars.
- **CameraX 1.4.2** - camera-camera2, lifecycle, video and view connect the camera, preview and rolling video recording.
- **ML Kit Text Recognition 16.0.1** - Reads scoresheet images on the phone; the user reviews the text before importing statistics.
- **Vosk Android 0.3.47** - Transcribes the verified audio into Indian-English scoring commands offline.
- **Sherpa-ONNX 1.13.8** - Runs the bundled ERes2Net speaker model to compare incoming speech with the enrolled official.
- **JNA 5.13.0** - Provides native-library access required by the speech stack.
- **Bundled models** - speaker.onnx identifies speaker similarity; speech.zip contains the Vosk small Indian-English model.
- **Neon serverless driver** - @neondatabase/serverless connects the Vercel JavaScript API to cloud PostgreSQL.
- **JUnit 4.13.2** - Checks scoring rules, replay/undo behavior and voice-command parsing in unit tests.
- **AndroidX Test / Espresso** - Runs device integration checks and interacts with screen controls.
- **Declared UI dependencies** - Material 1.14.0, Activity KTX 1.13.0 and ConstraintLayout 2.2.2 are included; current screens mostly use Java-created native widgets.

Widgets and layout controls

- **TextView** - Displays the score, headings, commentary, statuses and player statistics.
- **Button** - Triggers actions such as logging runs, opening tools and checking a replay.
- **EditText** - Accepts player names, PINs, match settings, corrections and reviewed OCR text.
- **Spinner + ArrayAdapter** - Shows selectable lists such as teams, players and rule options.
- **CheckBox** - Enables optional rules and the voice wake-word setting.
- **ProgressBar** - Displays the horizontal statistical win-probability estimate.
- **LinearLayout** - Arranges controls in vertical forms and horizontal scoring/navigation rows.
- **ScrollView** - Makes long forms, scorecards and tool screens scrollable.
- **AlertDialog** - Shows input forms, confirmation prompts and the cancellation window for major corrections.
- **ImageView** - Previews the selected scoresheet photograph in the scanner.
- **PreviewView** - Shows the CameraX live camera feed in stump-cam mode.
- **VideoView** - Plays the exported review clip at reduced playback speed.
- **WebView + WebViewClient** - Opens the public spectator scoreboard and restricts navigation to the approved site.
- **Custom FieldView** - Draws the cricket ground and lets the captain drag nine fielder markers.
- **Canvas, Paint, GradientDrawable** - Draw the field and style native UI cards, colors and rounded backgrounds.

Listeners, callbacks and Android methods

- **setOnClickListener()** - Runs code when a control is tapped, such as saving four runs or starting voice capture.
- **setOnLongClickListener()** - Opens detailed delivery entry when a run button is held down.
- **setOnCheckedChangeListener()** - Saves the optional wake-word setting when its checkbox changes.
- **setOnShowListener()** - Attaches validation to a dialog's buttons after the dialog becomes visible.
- **setOnCancelListener()** - Resets voice-capture state when the command-confirmation dialog is cancelled.
- **setOnPreparedListener()** - Configures the replay's media player after the video is ready.
- **onTouchEvent(MotionEvent)** - Detects touches and dragging on the field-placement map.
- **onDraw(Canvas)** - Redraws the cricket ground and current fielder positions.
- **onCreate(Bundle)** - Initializes an Activity, its views and the resources needed by that screen.
- **onDestroy()** - Releases camera, audio, native speech models and temporary replay resources.
- **Intent / activity result** - Opens another screen or the system image picker and returns selected data.
- **Handler / main thread** - Schedules UI work and the three-second cancellable correction confirmation.
- **ExecutorService** - Runs expensive speech/model work away from the UI thread.
- **Runtime permissions** - Requests microphone and camera access before recording audio or video.

Code structure: where each responsibility lives

- **MainActivity.java** - Builds Home, Score, Cups, Teams and Tools; connects user actions to match operations.
- **domain/Cricket.java** - Defines rules, teams, matches, innings, events, statistics and tournament progression.
- **domain/VoiceCommands.java** - Converts recognized words into scoring, review and correction commands.
- **data/Repository.java** - Stores records in boundary.db using SQLiteOpenHelper and maintains the publication outbox.
- **data/JournalCodec.java** - Serializes, compresses and encrypts the private match journal for recovery.
- **Publication.java** - Publishes queued snapshots and retries through a network-aware Android JobService.
- **OfficialAccess.java** - Checks the local official PIN, session timeout and failed-attempt limits.
- **VoiceActivity / ScannerActivity** - Handle speaker enrollment and scoring audio, or OCR review and scorebook import.
- **StumpCamActivity / ReplayBuffer / ReplayServer** - Capture bounded camera segments, export a replay and serve paired local review.
- **FieldView / UiInsets / SpectatorActivity** - Provide the interactive field, safe screen padding and public-score viewer.
- **api/matches.js** - Checks match authority and revisions, then reads or writes PostgreSQL data.
- **web/index.html** - Renders public live matches and scorecards; refreshes from the API every three seconds.
- **AndroidManifest.xml / res/** - Declare screens, permissions and services; store themes, strings, icons and backup rules.
- **Gradle / package.json / vercel.json** - Configure Android dependencies and signing, backend dependencies and web deployment.

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Project terms and data flow

- **Class / object / enum** - A class defines a type, an object holds its data, and an enum restricts choices such as delivery-extra type.
- **Inheritance / override** - Activities extend AppCompatActivity; overridden lifecycle methods customize Android behavior.
- **Lambda / callback** - A short function such as v -> guard(action) runs later when a UI or asynchronous event occurs.
- **Domain layer** - Framework-independent cricket logic calculates scores consistently without depending on screen widgets.
- **Event journal / replay** - Stores match actions and rebuilds state by applying them in order, enabling undo and historical corrections.
- **Offline-first / outbox** - Saves on the phone first, then keeps pending publication work until a network transfer succeeds.
- **SQLite transaction** - Commits the local match and its pending publication together to avoid partially saved changes.
- **API / HTTPS / JSON** - The Android app and website exchange structured score data with the Vercel backend over encrypted transport.
- **Neon PostgreSQL** - Cloud database holding public snapshots and separately encrypted private match journals.
- **Revision / idempotency** - Version checks reject stale/conflicting writes; an identical retry can succeed without changing the result.
- **AES-GCM / PBKDF2** - AES-GCM protects private cloud journals; salted, iterated PBKDF2 hashes protect the local PIN.
- **JobScheduler / backoff** - Schedules network-required sync and increases retry delay after failures.
- **OCR / embedding / heuristic** - OCR reads image text, embeddings compare voices, and local heuristics estimate tactics, win chances and revised targets.
- **Production boundaries** - Shared account roles are not implemented; voice accuracy needs field validation; rain targets are not official DLS.